

# How Inflation Ends Is How Matter Generation Begins: An N-Body Embedding of the Universal Interaction with No Timing Rule, and an End-to-End Genesis Run

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## Abstract

Two preceding papers each supplied half of the story. The onset-mode paper [2] gave the way a universe begins (only unstable self-consistent closures undergo latent exponential amplification, ending in a three-direction metastable structure) — but there the dynamics stops and there is no matter. The two-body paper [1] gave the generation mechanism (a universal inelastic vertex whose only input is the waveforms) — but its collisions were loop counters, with when and where unanswered. This paper closes the gap **without introducing any timing rule**. Embedding the two-body vertex into the N-body relational wave system (the  $M = N(N - 1)/2$  relational waves of the complete graph  $\times$  a multi-slice register axis) with the interaction always on, we measure the following. (1) The apparent epoch of generation is set by a **two-factor law** — generation rate = condensation gate  $\times Cf^2$  (gate ratio 2.74 measured). And the condensation is precisely the endpoint of the onset paper: **that the ending of the previous paper is the opening condition of this one is a measurement, not an assumption**. (2) The two-body laws hold unchanged in the many-body system: the ignition law transplants with power  $p = 2.0011$  and coefficient constant to 0.4%, and the pair-structure census holds up to the hair (winding-number) complete version (sum-rule exclusivity ratio  $10^{289}$ , hair lock 0.9997). (3) Running vacuum  $\rightarrow$  latent exponential amplification  $\rightarrow$  three-direction condensation  $\rightarrow$  matter precipitation under a single dynamics with no switches (GENESIS), we discover **seed unification**: the perturbation that ignites inflation and the seed that becomes matter are one and the same ( $\varepsilon$ -law agreement to 2%). (4) Locality does not live on the graph — concentration onto the complete graph leaves the generation rate unchanged (ratios 1.28, 1.02) — but on the register axis (response  $\times \sim 19.6$ ). Products are pinned at their birth position, and harmonic-closure dispersion provides shape-invariant translation (1D deviation below  $3 \times 10^{-4}$  cells; 3D deviation 0.0000). (5) The generation direction is a phase lottery of the seed ( $P(+) = 0.52$  over 50 random phases), and net matter arises by rectification through the  $Cf^2$  autocatalysis (80% of the ensemble grows net). As conditional derivations we present the mediated-vertex uniqueness theorem (closure conservation forces  $g_2 = -g_1$  **and** pair symmetry of the mediated strengths — the two-body uniqueness is its  $M = 2$  special case) and the identity of the computational structure (the adjacency sum folds exactly into two edge $\rightarrow$ node $\rightarrow$ edge passes; the operation count is linear in the state size, the theoretical lower bound, and what guarantees this is anonymity itself). We report with assumptions, conditional derivations, and measurements strictly separated, and we measure the value of the assumption by the demonstrated connection — that the physics of two independently published papers reproduces simultaneously as different chapters of a single dynamics with no switches.

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## 1. The Character of This Paper — Separating Assumption, Derivation, and Measurement

This paper neither proposes a new interaction nor proves one correct. It takes the universal inelastic vertex — posited as an explicit assumption and conditionally unified in the two-body paper [1] — embeds it into the N-body relational wave system with no timing rule whatsoever, and measures what happens.

- **(a) Assumptions:** the inheritance of the vertex from the two-body paper, and the design choices of the embedding (Section 4).
- **(b) Conditional derivations:** the mediated-vertex uniqueness theorem and the identity of the computational structure (Section 5).
- **(c) Measurements:** the resolution of timing, the transplantation of the two-body laws, the end-to-end genesis run, position and motion, and the direction lottery with rectification (Sections 6–10).

What the reader can verify are (b) and (c); (a) is the starting point.

## 2. Claims

**First — the motivation and design principle of this paper — we do not choose by hand when the interaction happens.** The two preceding papers each gave half. The onset paper gave the way a universe begins, but the dynamics stops at its endpoint (the three-direction metastable structure) with no matter. The two-body paper gave the generation mechanism, but its collisions were loop counters. Filling this gap with a rule such as “invoke the interaction after condensation” would be an IF statement, and the anonymity the two-body paper preserved would collapse in the many-body setting. The design of this paper is single: **the interaction acts at all times.** Eventness is an output, not an input.

**Second, we enumerate all assumptions (Section 4).** (i) The linear dynamics is inherited unmodified from Papers 7 and 8 (read-only import, SHA recorded). (ii) The vertex extends the unquified two-body form to an adjacency sum — each relational wave couples through the simple sum of two-wave vertices with the relational waves sharing a body (no vertex of three or more waves). (iii) The state is a multi-slice register  $C \in \mathbb{C}^{M \times N_{reg}}$  with a shared readout of the combined spectrum. (iv) No new constants — the strength reuses the readout  $R$ . **The four items are design choices, not derivations.**

**Third, the mediated-vertex uniqueness theorem (conditional derivation, Section 5.1).** Within the assumed class, closure conservation forces  $g_2 = -g_1$  and pair symmetry of the mediated strengths. The two-body  $g_2 = -g_1$  is the  $M = 2$  special case of this theorem. What closure demands of the strengths is pair symmetry only; the arithmetic-mean rule is an additional experimental fact not derived from closure. Extension theorem T-L: register-point-dependent pair-symmetric strengths  $R_{ee'}(n)$  also conserve closure exactly — locally varying interaction strength is compatible with the axiom (machine verified).

**Fourth, the identity of the computational structure (conditional derivation, Section 5.2).** The adjacency sum of the vertex folds exactly into a two-pass edge→node→edge aggregation (vertex moments and self-subtraction). The operation count is linear in the state size,  $O(M \cdot N_{reg} \log N_{reg})$  — no all-pairs checks  $O(N^4)$ , no adjacency scans  $O(N^3)$  — reaching the theoretical lower bound. What makes this folding possible is anonymity itself (one law for all pairs, no pair-specific constants).

**Fifth, Measurement 1: the timing problem resolves into the two-factor law (Section 6).** Generation rate = condensation gate  $\times Cf^2$ . The measured gate ratio is 2.74. Condensation is the endpoint of the onset paper — the three-direction metastable structure. That the previous paper’s ending is this paper’s opening condition is a measurement, not an assumption.

**Sixth, Measurement 2: the two-body laws hold unchanged in the many-body system (Section 7).** The ignition law reproduces with power  $p = 2.0011$  and constant coefficient (0.4%). The census holds up to the hair-complete version: sum-rule exclusivity ( $10^{-289}$  for unpredicted bins), hair lock 0.9997, simultaneous rise from exact zero.

**Seventh, Measurement 3: the end-to-end genesis run GENESIS (Section 8).** Vacuum  $\rightarrow$  latent exponential amplification  $\rightarrow$  three-direction condensation  $\rightarrow$  matter precipitation under one dynamics with no switches. Discovery: **seed unification** — the perturbation that ignites inflation and the seed that becomes matter are one and the same.

**Eighth, Measurement 4: position and motion (Section 9).** Locality does not live on the graph (concentration ineffective; ratios 1.28, 1.02). Locality lives on the register axis (response  $\times \sim 19.6$ ). Products are pinned at their birth position, and motion is provided by the harmonic-closure dispersion  $\omega_k = k\omega_1$  [6] (shape-invariant translation demonstrated in 1D/3D).

**Ninth, Measurement 5: the direction lottery and rectification (Section 10).** The generation direction is set by the seed’s phases ( $P(+) = 0.52$  over 50 random phases). The instantaneous sign predictor scores 0.70 — above chance (0.5) but short of the preregistered criterion 0.9; the determinism of the lottery is only partially confirmed (a complete instantaneous predictor is an open problem). Net matter arises not from a hand-placed symmetry breaking but from rectification by the  $Cf^2$  autocatalysis; 80% of the ensemble grew net. This is a measurement of a mechanism inside the model; we do not claim it explains the matter excess of the real universe.

**Tenth, we record a null result (Section 11).** The selectivity of the E8-related special values ( $N_{reg} = 124, 137, 144, 248$ , etc.) is null on every instrument tested.

**Eleventh, we record the full process of refutation and correction (Section 12),** including three families of instrument error (conservation criteria, generation-rate predictors, the motion centroid gauge).

Finally, what this paper does not claim, and the falsification conditions (Section 15).

### 3. Motivation — the Two Papers Were Not Connected

The conclusion of the onset paper [2] is this: only an unstable self-consistent closure (a single coherent condensate) undergoes latent exponential amplification, and nothing amplifies out of white chaos. The endpoint of amplification is the three-direction metastable structure, where the linear dynamics completes. **Spacetime was made. But there is no matter and nothing happens.**

The conclusion of the two-body paper [1] is this: a single IF-free map seeing only waveforms generates fermionic structure — induced, autocatalytic, pair-correlated. **We know how to make matter. But there is no universe.** The two-body “collision” was a loop counter; when, where, and why collisions occur lay outside the dynamics.

This gap was not an incompleteness but a candidate fatal wound. Choosing the timing by hand — “invoke the interaction after inflation” — is an IF statement of the same rank as discriminating particle species, and the anonymity the two-body paper defended would collapse at the moment of embedding. A model that invokes its interaction only when convenient has no standing to speak about generation. Hence there is only one design: embed with the interaction always on, and measure whether “when” is output as physics. That is this paper.

## 4. Model and Assumptions

### 4.1 The relational wave graph (inherited)

The fundamental variables do not live on bodies (particles). Each pair of the  $N$  bodies (nodes) carries one relational wave  $z_e \in \mathbb{C}$  (an edge;  $M = N(N-1)/2$  in total). Nodes themselves hold no variables; they are aggregation points for ledgers. Two relational waves are eligible to interact only when they share a body (adjacency) — the shared body mediates (Figure 1). The linear dynamics (generator  $K_{ef} = \sin(\theta_f - \theta_e)$ , Cayley step; self-consistent closure, latency, burst, three-direction generation) is inherited unmodified from the original code of Papers 7 and 8 [3,4] (SHA-256 recorded, read-only import).

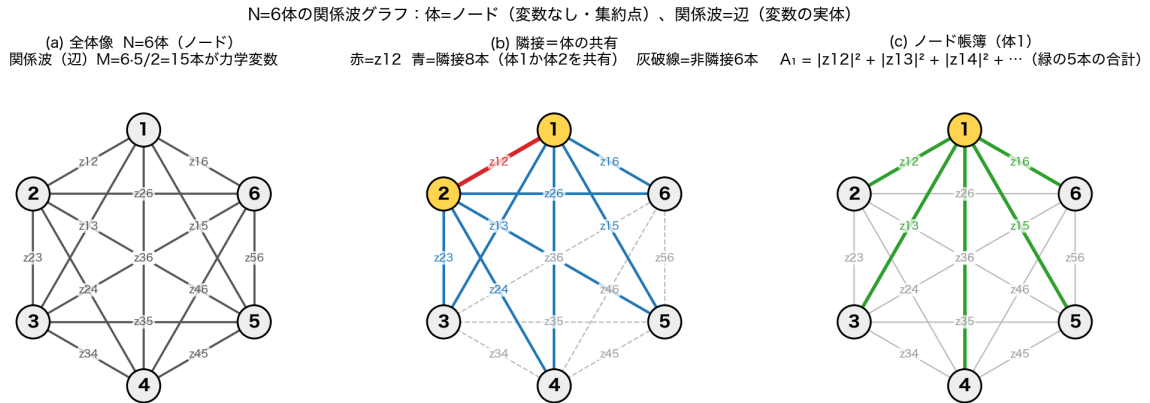


Figure 1: Figure 1 relational wave graph

**Figure 1:** the relational wave graph at  $N=6$ . (a) Bodies = nodes; relational waves = edges ( $M=15$  dynamical variables). (b) Adjacency = sharing a body: the interaction partners of  $z_{12}$  (red) are only the 8 edges sharing body 1 or body 2 (blue); the 6 edges sharing no body (gray dashed) do not couple directly. (c) The node ledger: the sum  $A_1$  of the relational waves meeting body 1.

### 4.2 Assumptions of the embedding

| #     | Assumption                                  | Content                                                                                                                                                                                                                        |
|-------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (i)   | Inherited linear dynamics                   | Read-only import of the Papers 7/8 originals. Acts independently on each slice (register harmonic)                                                                                                                             |
| (ii)  | Vertex = adjacency sum of two-wave vertices | Each edge $e$ couples through the simple sum over adjacent edges:<br>$\delta z_e = i \sum_{e' \sim e} R_{ee'} ( z_{e'} ^2 z_e - z_{e'}^2 \bar{z}_e)$ . No vertex of three or more waves                                        |
| (iii) | Multi-slice register                        | State $C \in \mathbb{C}^{M \times N_{reg}}$ . The vertex acts pointwise in register position space. The readout is shared over the combined spectrum:<br>$\theta = \text{atan2}(\sqrt{P_f}, \sqrt{P_b})$ , $R = \sin^2 \theta$ |
| (iv)  | No new constants                            | The strength reuses $R$ . No discriminating branches (the implementation's $r > 0$ is a zero-strength numerical skip)                                                                                                          |

The four items are design choices. (ii) two-wave and (iv) anonymity are direct inheritances of the two-body assumptions; (i) and (iii) are embedding choices specific to this paper (Figure 2).

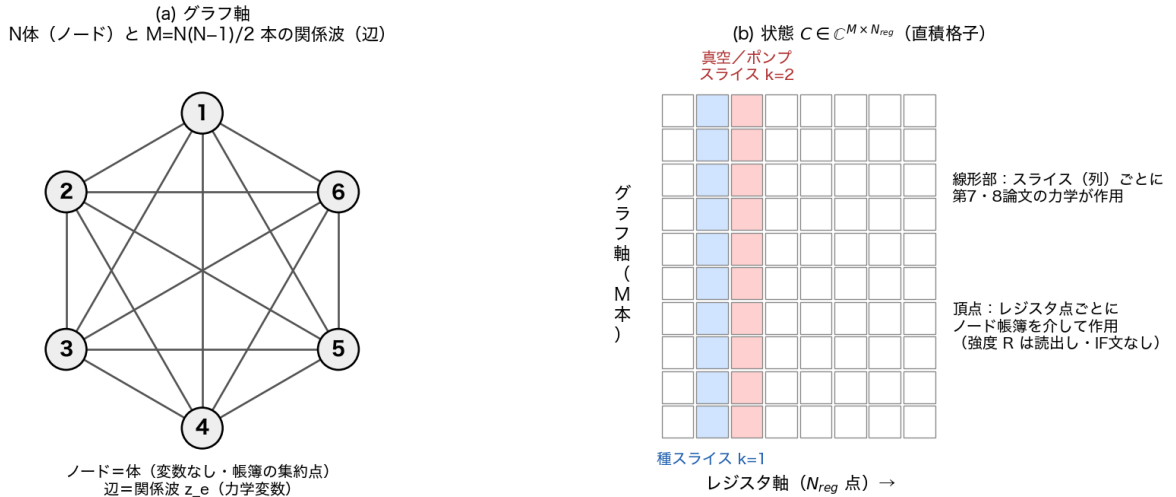


Figure 2: Figure 2 architecture

**Figure 2:** the state is the product lattice of the graph axis ( $M$  relational waves)  $\times$  the register axis ( $N_{reg}$  points). The linear part acts per slice (column); the vertex acts per register point through the node ledgers.

## 5. Conditional Derivations

### 5.1 The mediated-vertex uniqueness theorem

For the most general form in the assumed class (two-wave, pointwise, cubic, common-phase invariant, no self-scattering),  $\delta z_e = i \sum_{e' \sim e} (g_1 R_{ee'} |z_{e'}|^2 z_e + g_2 R'_{ee'} z_{e'}^2 \bar{z}_e)$ , conservation of the zero-square-sum closure  $\sum_e z_e^2$  (per register point) forces:

$$g_2 = -g_1 \quad \text{and} \quad R_{ee'} = R_{e'e} \text{ (pair symmetry).}$$

The proof is by pairwise cancellation of the first variation: the contribution of the edge pair  $(e, e')$  is  $g_1 R_{ee'} |z_{e'}|^2 z_e^2 + g_2 R'_{ee'} z_{e'}^2 |z_e|^2$  plus its  $e \leftrightarrow e'$  exchange, and it vanishes identically only when the coefficients are opposite and the strengths pair-symmetric. Three points. First, this is uniqueness within

the class. Second, the two-body  $g_2 = -g_1$  [1] is the  $M = 2$  special case (the adjacency sum has one partner). Third, closure demands of the strengths **pair symmetry only**: as the mixing rule when two edges have different  $R$ , both the arithmetic mean and the product conserve closure exactly (machine test T5). That the actual dynamics is consistent with the arithmetic mean is an additional experimental fact.

**Extension theorem T-L**: since pair symmetry holds per point  $n$ , register-point-dependent pair-symmetric strengths  $R_{ee'}(n)$  also conserve closure exactly. Machine check: drift  $1.0 \times 10^{-17}$  (machine zero) for random  $n$ -dependent pair-symmetric  $R$ ;  $1.7 \times 10^{-2}$  (broken) when asymmetric. **Locally varying interaction strength is compatible with the axiom** — the axiom does not forbid locality.

## 5.2 The identity of the computational structure — anonymity guarantees computability

Naively, evaluating the adjacency sum scans  $2(N - 2)$  adjacent edges per edge, costing  $O(M \cdot N) = O(N^3)$  per register point. But by (ii) two-wave additivity (no cross terms between partners) and the separability of the pair-symmetric mean strength, the adjacency sum folds **exactly**: for edge  $e = (u, v)$ ,

$$\sum_{e' \sim e} |z_{e'}|^2 = A_u + A_v - 2|z_e|^2, \quad A_v \equiv \sum_{e \ni v} |z_e|^2$$

(likewise for the four ledgers including the  $z^2$ -type and  $R$ -weighted ones). The computation is two passes — “edges  $\rightarrow$  node aggregation” and “nodes  $\rightarrow$  edge read-back with self-subtraction” — each  $O(M)$ . This is an identity, not a mean-field approximation; agreement with the naive implementation is verified to machine precision. The total per step is  $O(M \cdot N_{reg} \log N_{reg})$  — the same order as the trivial lower bound in the RAM model of having to read the whole state once per step, and in this sense the theoretical lower bound.

The physical meaning: information flows edge  $\rightarrow$  shared node  $\rightarrow$  edge. **The interaction is not a pairwise direct check but aggregation and redistribution mediated by the bodies** — the  $O(M)$  algorithm is the literal implementation of “mediated vertex.” And what makes the folding possible is anonymity itself: give each pair its own coupling table  $R_{ee'}$  (violating anonymity) and separability dies, returning the cost to  $O(N^3)$ . A physical principle directly guarantees a computational property.

**Measured conserved quantities** (under the discipline of deriving the invariant before judging): relative norm drift  $3.1 \times 10^{-14}$ ; vertex-only pointwise closure drift  $7.3 \times 10^{-11}$ ; linear-only slice closure drift  $1.4 \times 10^{-15}$  (each judged only where it is an invariant; see Section 12). The ON/OFF separation of generated content is 11 orders; the ON amount is consistent with the value derived from the ignition law within a factor 4.2.

## 6. Measurement 1: the Two-Factor Law — Resolving the Timing Problem

With the always-on embedding, “when generation happens” did not vanish; it reappeared as physics:

$$\text{generation rate} = \text{condensation gate} \times Cf^2.$$

**The condensation gate** (Figure 3a): in a control with identical norm and identical seed, with the pump placed either in the condensed state (the three-direction metastable structure) or in a non-condensed state (a non-coherent state of the same energy), the generation rates are  $6.30 \times 10^{-8}$  vs.  $2.30 \times 10^{-8}$  — **gate ratio 2.74**. Condensation is precisely the endpoint of the onset paper [2].

**Era modulation** (Figure 3b): the windowed generation rate of the end-to-end run (Section 8) opens monotonically — latency  $3.3 \times 10^{-7} \rightarrow$  burst  $1.4 \times 10^{-6} \rightarrow$  post-condensation  $6.4 \times 10^{-6}$  — a factor  $\times \sim 19.5$  from latency to post-condensation. The interaction acts at all times, yet the apparent epoch of generation is made by the age of the universe.

**Figure 3:** (a) the condensation-gate control. (b) era modulation of the end-to-end run. No timing rule was written; timing was output.

**Reading (explicitly an interpretation):** this measurement suggests that it is not that “after inflation ends, matter generation begins as a separate phenomenon.” The end of expansion (linear side: saturation of the unstable growth, settling into the three-direction structure) and the onset of generation (nonlinear side: opening of the condensation gate) can be read as **two observables of the same condensation**

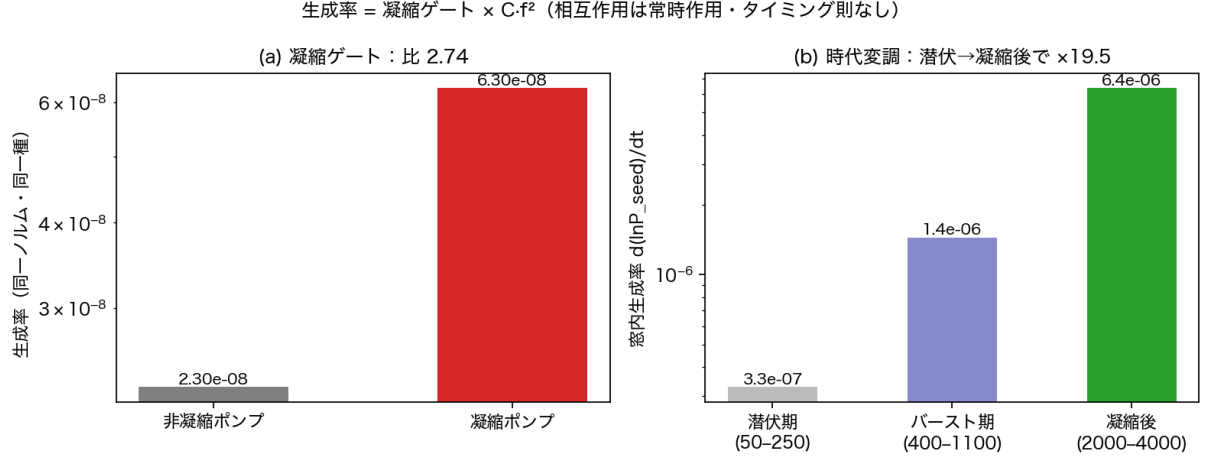


Figure 3: Figure 3 two-factor gate

**transition** — spacetime formation and matter generation are not two serial processes but the outer geometric face and the inner generative face of one transition. The reading is an interpretation; the two quantities supporting it (gate ratio, era modulation) are measurements.

## 7. Measurement 2: Transplantation Tests of the Two-Body Laws

**The ignition law** (Figure 4): over a four-decade sweep of the seed fraction  $f_0$ , the initial generation rate obeys rate =  $C f^p$  with  $p = 2.0011$  and a coefficient spread of 0.37%. Across an engine revision (stage 2 → stage 3 shared-readout), the law is unchanged:  $p = 2.0015$ ,  $C = 5.88$ . The two-body autocatalytic ignition law [1] keeps its form inside the many-body sea.

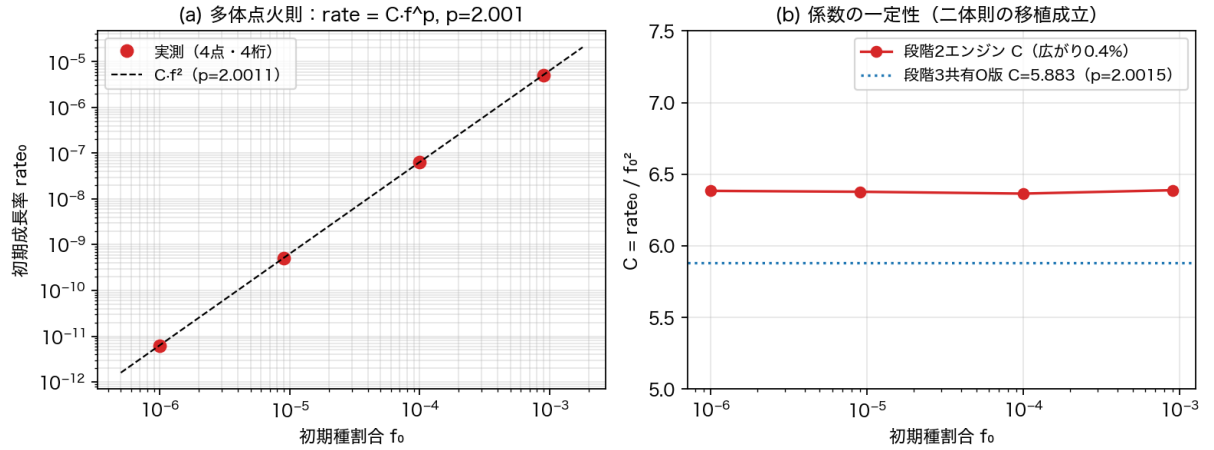


Figure 4: Figure 4 ignition transplant

**Figure 4:** (a) the many-body ignition law rate =  $C \cdot f^p$  (log-log, four decades). (b) constancy of the coefficient.

**The pair-structure census, hair-complete** (Figure 5): the preregistered census with narrow-band pumps and a single seed was run on the two-dimensional ledger of register ( $k$ ) and hair ( $\eta$  winding  $m$ ). The hair-resolved power of the partner band  $k = 3$  appears only at the hair dictated by the sum rule  $m^* = 2m_B - m_s = 3$  (**exclusivity ratio**  $6.1 \times 10^{289}$  against unpredicted hairs at machine zero), with pair-correlation lock 0.9997. The partner band rises from exact zero simultaneously with the sidebands. The ledger is machine-derived from a single vertex application — nothing hand-set. The pair-correlation lock decays slowly over long times (0.988 at the tenth window; Figure 5b) — the cause is unresolved and recorded as an open problem.

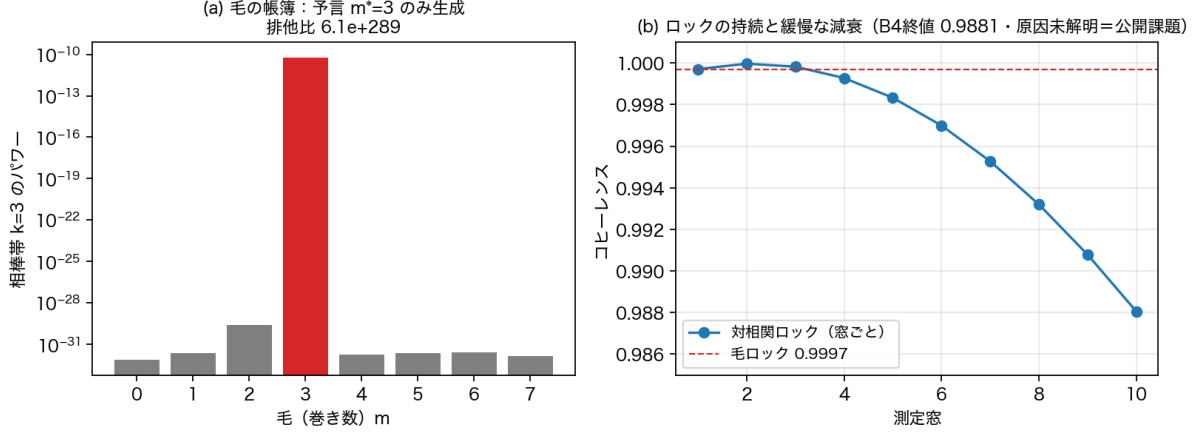


Figure 5: Figure 5 hair-complete census

**Figure 5:** (a) the hair ledger: only the predicted  $m^* = 3$  generates. (b) persistence and slow decay of the lock (open problem).

## 8. Measurement 3: GENESIS — the End-to-End Genesis Run

From an initial state of vacuum (the unstable branch of the two-chirality vacuum pair) plus a tiny seed  $\delta$ , we ran the single dynamics for 4000 steps (Figure 6). There are no switches, no era changes, no parameter changes.

**Figure 6:** top: the slice-2 vacuum’s unstable branch amplifies exponentially after latency (crossing  $t = 180$ ) and condenses into the three-direction metastable structure. Middle: the one-step residual  $r$  (clock). Bottom: the moving-window generation rate opens with the era ( $\times \sim 19.5$ ) — the end-to-end demonstration of the two-factor law.

**Discovery 1: seed unification.** The pre-run prediction assumed inflation would be ignited by the system’s intrinsic floor (the parent eigenmode residual  $2.1 \times 10^{-13}$  [2]) and registered the crossing at 1161–1171. The measurement was 555 (the  $\delta = 10^{-6}$  series) — far outside the band. **Diagnosis: the seed  $\delta$  injected for matter was itself igniting inflation.** Substituting  $\varepsilon = \delta = 10^{-6}$  into the onset paper’s  $\varepsilon$ -law (crossing linear in  $-\ln \varepsilon$  with slope  $1/\lambda$ ) predicts about 543, agreeing with the measured 555 to 2%. The seed that ignites inflation and the seed that becomes matter are not separate devices but one and the same perturbation. A failed prediction turned into the discovery of the connection. Figure 10 summarizes the causal structure up to this point.

**Figure 10:** the causal diagram of the genesis run. From the single perturbation  $\delta$ , two branches: the geometry branch (unstable departure  $\rightarrow$  latent exponential amplification  $\rightarrow$  three-direction condensation  $\rightarrow$  gate opens) and the seed branch (odd-parity seed  $f_0 \rightarrow$  autocatalytic factor  $Cf^2$ ). They merge at generation rate  $= G_{cond} \times Cf^2$ , followed by matter precipitation, localization, motion, and pair creation. There is no timing rule anywhere.

**Consequence: a sea with a small seed is effectively inert.** At  $\delta = 10^{-6}$  ( $f_0 = 10^{-12}$ ) generation over 3000 steps stays below the measurement limit ( $|g| < 10^{-11}$ ), while at  $\delta = 10^{-2}$  it is clear ( $6.4 \times 10^{-6}$ ). The two-body consequence — ignition time scaling as  $1/f_0^2$  — appears unchanged in the many-body universe.

## 9. Measurement 4: Position and Motion

**Locality does not live on the graph** (Figure 7a). Concentrating the seed onto a star subset of the complete graph (concentration  $\times \sim 6$ ) leaves the generation rate unchanged from a uniform seed (ratios 1.28, 1.02). The complete graph  $K_N$  has no distance, and the vertex sees adjacency only through node-ledger aggregation, so “closeness” on the graph does not exist in the physics. Stated precisely, this is not a permanent theorem but the measurement that “in the present complete-graph, anonymous-adjacency-sum embedding, graph concentration did not change the generation rate” — the claim of this paper is that the primitive complete graph is not position space, and the possibility remains open that an induced distance

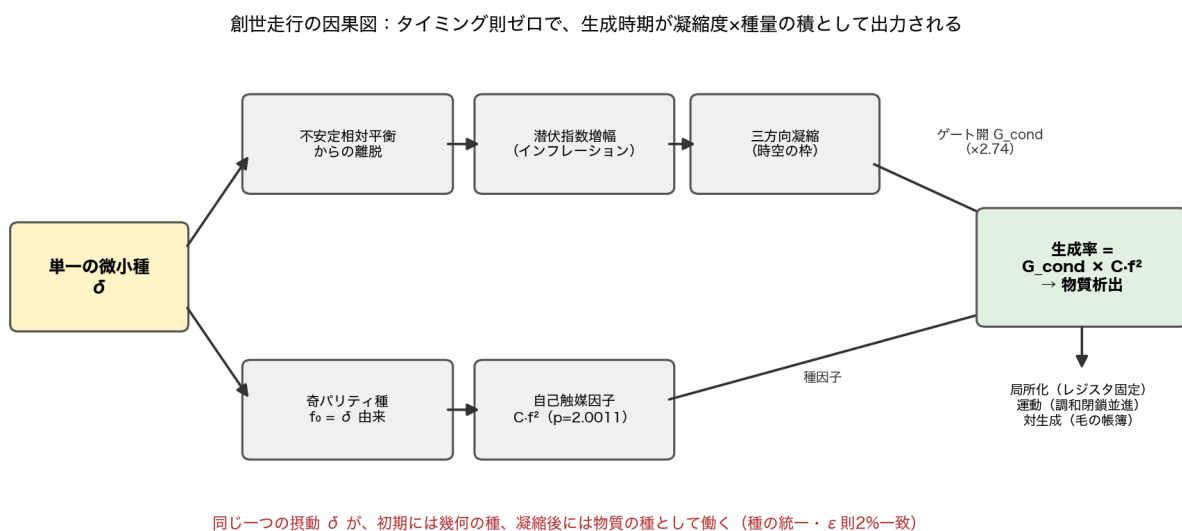
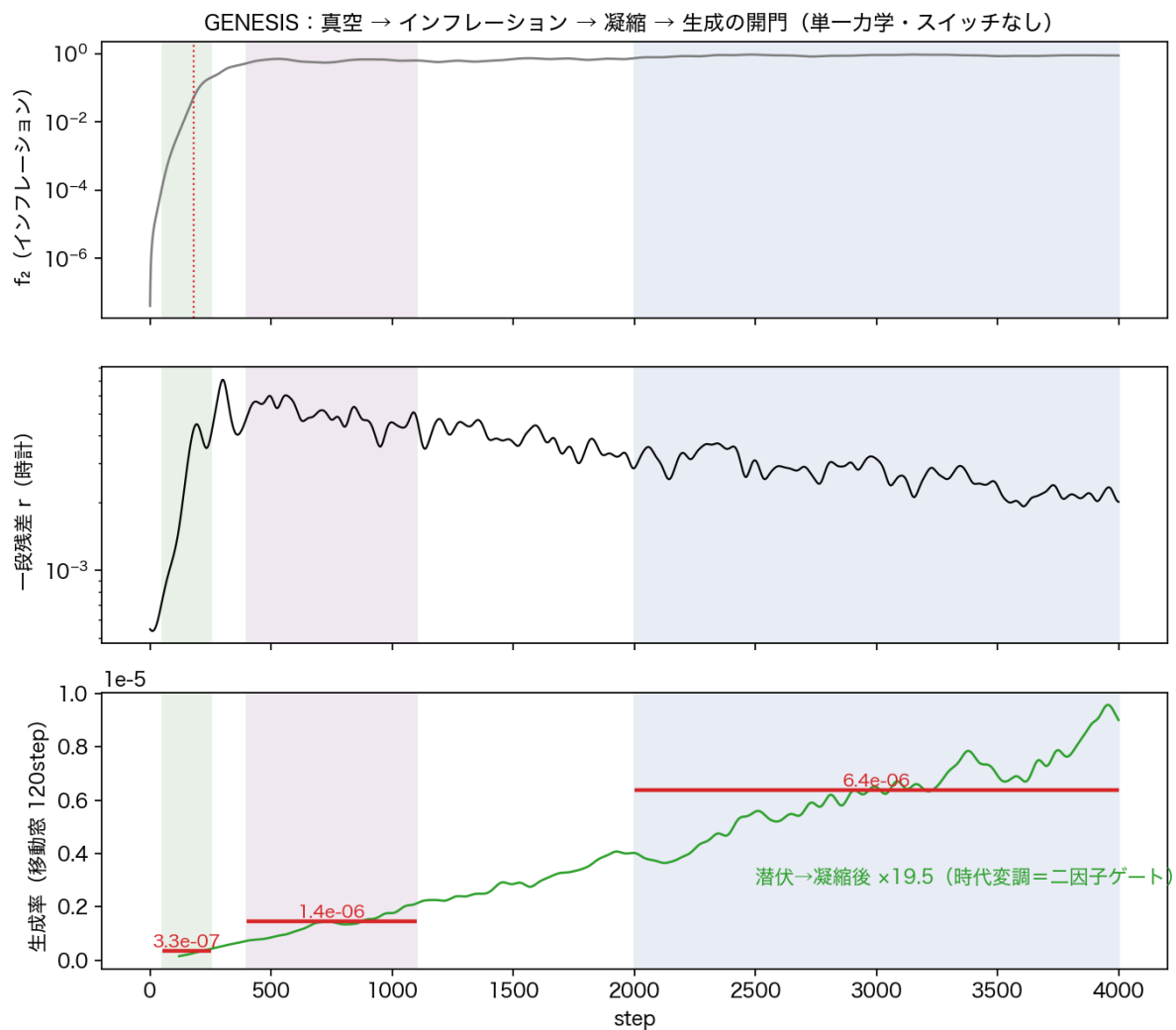


Figure 7: Figure 10 causal diagram of the genesis run



structure appears on the graph after the unification of the three-direction frame with the register (Section 14-4).

**Locality lives on the register axis** (Figure 7b). Localizing a seed of identical power into a packet in register position space (concentration  $\times \sim 8$ , effective width  $PR=2$ ) raises the generation response  $\times \sim 19.6$ . Moreover the register distribution of the products stays at  $PR=2$ , pinned at the birth position throughout growth — **particles have positions**.

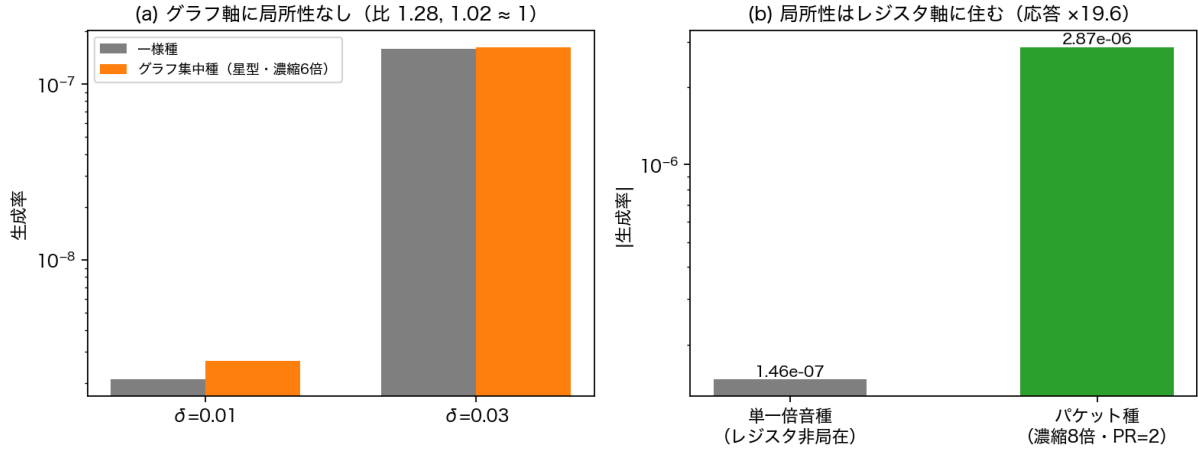


Figure 8: Figure 7 the address of locality

**Figure 7:** (a) graph concentration is ineffective. (b) register concentration gives  $\times \sim 19.6$ . The address of locality is the register axis, not the graph.

**Motion** (Figure 8). The linear part has no dispersion (all harmonics equal speed), so a pinned packet does not move. Introducing the harmonic-closure dispersion  $\omega_k = k\omega_1$  — the dispersion relation the inverse-square-law paper [6] derived from the geometry of acceleration — as a register clock is nothing but a translation operator, and the packet moves rigidly at constant velocity. Measured: in 1D all 8 recorded points deviate from prediction by at most  $3 \times 10^{-4}$  cells, and at integer-cell shifts the waveform returns exactly to its initial shape ( $PR$  1.000). In 3D ( $8^3$  register, three independent velocities) all recorded points deviate by 0.0000,  $PR$  constant 3.39. This is rigid translation with the vertex ON and the seed present.

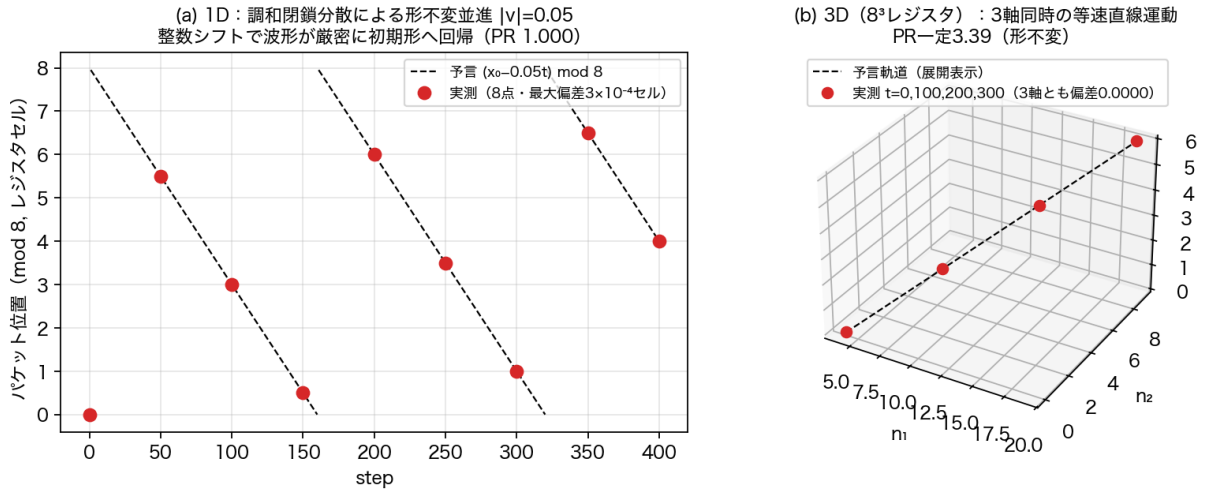


Figure 9: Figure 8 motion

**Figure 8:** (a) 1D translation (mod-8 display). (b) 3D uniform linear motion.

## 10. Measurement 5: the Direction Lottery and Rectification

**The direction lottery** (Figure 9a). The generation direction of an individual seed (matter side + or antimatter side −) is set by the seed’s phase profile. Over 50 random-phase packet seeds,  $P(+) = 0.52$  — very nearly a 50/50 lottery. The instantaneous sign predictor built from the odd projection of the vertex rate scores 0.70 — above chance but short of the preregistered 0.9. This shortfall is not evidence of randomness; it indicates that the present predictor’s observables likely fail to contain all the state information that determines the sign — candidates include hair-resolved phase alignment, interference among channels, and the projection sign onto the unstable eigenmode. Constructing a complete instantaneous predictor is an open problem (Section 14-2). In the channel-multiplicity test, the generation rate grows with the number of seed harmonics  $n_h$  with power  $q = 0.91$ , consistent with the reading that register locality (Section 9) is the Fourier dual of four-wave-mixing channel count.

**Ratchet rectification** (Figure 9b). If the direction is a 50/50 lottery, why does net matter remain? The autocatalysis  $Cf^2$  is a rectifier — the side that grows grows faster: over a random-phase ensemble of 20 runs  $\times$  800 steps, **80% grew net** (mean final ratio 1.0013). Net content arises with no hand-placed symmetry breaking, from reversible elementary processes plus autocatalysis alone. This is a measurement of a mechanism inside the model; we do not claim it explains the matter excess of the real universe.

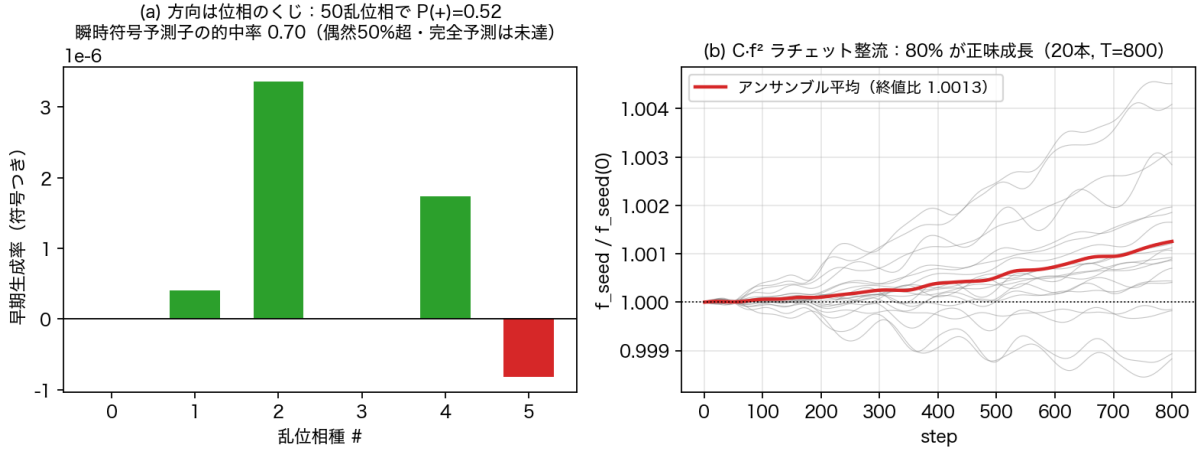


Figure 10: Figure 9 lottery and ratchet

**Figure 9:** (a) signed generation rates (six random-phase examples) and the 50-seed statistics. (b) ensemble trajectories and their mean (red).

## 11. Recording a Null Result — E8 Special-Value Selectivity

We tested whether register counts  $N_{reg}$  at E8-related special values (124, 137, 144, 248) behave selectively, using four instrument families — all-band linear phases, first-order ignition rates, thermalization entropy, and ladder-front collision interference (Nyquist-band power). **All null** (every special-to-neighbor ratio inside the preregistered band  $[0.5, 2]$ ). We record the motivation (an internal E8 structural hypothesis of the series), the instrument designs, and the failed early instruments (Section 12). Establishing that selectivity is absent deserves the same weight as claiming it is present.

## 12. Record of Refutations and Corrections

| # | Error                                                                                                                                                                                | Discovery  | Correction                                                                                                                                                               |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Conservation criteria<br>measured non-invariants<br>(pointwise closure is not<br>invariant under<br>slice-independent linear<br>dynamics) with an<br>unjustified threshold $10^{-6}$ | U2/U3 FAIL | Derive the invariant first,<br>then judge (all pass in v2).<br>Discipline: <b>always derive<br/>the invariant before<br/>preregistering a<br/>conservation criterion</b> |

| # | Error                                                                                                                                                                                        | Discovery                                                     | Correction                                                                                                                                                                                                                                                |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 | The same error recurred (slice closure is not invariant under vertex ON)                                                                                                                     | Stage-3 B2 FAIL                                               | Re-judged on the pointwise field closure (the true invariant); discipline written out                                                                                                                                                                     |
| 3 | Generation-rate predictors wrong twice (cumulative vs. instantaneous; own-edge product vs. mediated product)                                                                                 | D1-P2 FAIL                                                    | The two-channel complete predictor (parametric + beat-XPM) — stepwise coherence 0.9998                                                                                                                                                                    |
| 4 | Blind spots of the D2 instrument (ignition window closed to low slices; the negative-frequency ladder alias misread as wall arrival)                                                         | $t_{wrap} = 1$ for all $N_{reg}$                              | Third-generation ladder-front instrument (positive-branch restriction, front tracking) → null confirmed                                                                                                                                                   |
| 5 | Identically-zero defect of the motion gauge: for odd-harmonic packets ( $ W ^2$ has period $N/2$ ) the fundamental winding moment vanishes identically — the v1 position readings were noise | Found in figure inspection ( $ z_1  \sim 10^{-16}$ confirmed) | Corrected to the winding-2 moment (position defined mod $N/2$ ); additionally calibrated a sign error in the prediction ( $e^{+ikw_1}$ translates in the — direction). After correction: 1D all 8 points within $3 \times 10^{-4}$ ; 3D all points 0.0000 |

The three families (#1–2, #3, #5) are all the same mistake: building an instrument without deriving the quantity it measures. Having stepped on the same rake three times, we write the discipline out: **derive the invariance (or the predicted value) of the measured quantity before designing the instrument.**

### 13. Prior Work — Correspondences Found After Independent Construction

The architecture of this paper was constructed independently from the two axioms and the two-body vertex, not designed by relying on the following. Re-surveying after the design, we found these correspondences. They are supporting evidence that the framework is not an isolated fantasy; they are not proofs of correctness.

1. **Variables on edges — lattice gauge theory** [8]. In Wilson’s lattice gauge theory the gauge field (link variables) lives on edges, with interactions built through sites and plaquettes. Differences: a background lattice (fixed geometry) and a gauge group are inputs there, and matter fields live on sites; here there is no geometry (complete graph), groups are post-hoc classifications, and nodes carry no variables at all.
2. **Spacetime emergent from pairwise degrees of freedom — matrix models (IKKT/BFSS)** [9]. The fundamental degrees of freedom are matrices whose off-diagonal entries are strings connecting brane pairs (pairwise relational degrees of freedom); there is no ambient spacetime, and in Lorentzian IKKT Monte Carlo the nine-dimensional symmetry breaks spontaneously to an expanding three-dimensional space [10] — phenomenologically parallel to the three-direction emergence of this series. Differences: the action comes from superstring theory (supersymmetry as input); matrices carry diagonals (per-body position variables) whereas we are purely relational; and the address of locality differs (next item).
3. **Complete graphs and locality — quantum graphity** [11]. Identical to us up to the starting point — degrees of freedom on a complete graph with physics invariant under vertex permutations (= anonymity) — graphity proposed that locality emerges by the graph thinning out in a low-energy phase. Our measurement is **the opposite answer to the same question**: the graph stays complete, locality does not appear on it (Section 9), and it lives on the register axis.

4. **Edge oscillators with node mediation — topological Kuramoto** [12]. The recent framework placing phase oscillators on edges coupled through Hodge Laplacians (node sharing) is the closest mathematical form to our adjacency structure. Differences: their coupling is gradient-type and the subject is synchronization; ours is the cubic exchange flow selected by closure conservation, and the subject is generation.
5. **Exact reduction of all-to-all coupling — Kuramoto mean field** [13]. The order-parameter reduction  $O(N^2) \rightarrow O(N)$  is the classical precedent of our node ledgers. Differences: ours aggregates edge variables at nodes with self-subtraction (four ledgers including the anomalous  $z^2$ ), and the basis of exactness is anonymity (absence of pair-specific constants).
6. **End-to-end numerical genesis — preheating lattice codes** [14]. LATTICEEASY and its kin follow particle production from vacuum in one run. Differences: they input an expanding background, a potential, and field content. We have a single anonymous dynamics only; spacetime (the three-direction structure) is itself part of the output, and so is “when.”
7. **The relational-dynamics lineage — Barbour–Bertotti** [15]. The N-body implementation of Mach’s principle (dynamics on relative configuration space). Difference: there, particle positions remain the variables and relationalization is a quotient operation; here relations are the variables from the start and body variables appear nowhere in the theory.

## 14. Outlook — Open Problems (working guide; not claimed here)

1. **Order-parameterization of the condensation gate:** an analytic expression for what the gate ratio 2.74 measures. The condensed and non-condensed states are matched in norm, seed, and spectral occupation, so the difference must lie in phase correlations; the candidate is an adjacent-edge relative-phase coherence of the type  $\langle \text{Im}(\bar{z}_e z_{e'})^2 \rangle_{adj}$ . If extracted, the generation law closes as  $\Gamma = G(\mathcal{O}_{cond}) \cdot C f^2$ .
2. **Projection decomposition of seed unification:** what is shown so far is that one  $\delta$  ignited both. Whether  $\delta = \delta_{\parallel}$  (projection onto the unstable eigenspace) +  $\delta_{\perp}$  (projection onto the odd-parity generation channel) is one degree of freedom or two independent components is to be decided by the four-way perturbation experiment (eigen-direction on/off  $\times$  odd seed on/off). The complete instantaneous sign predictor (Section 10;  $0.70 \rightarrow 1$ ) belongs to the same projection decomposition.
3. **The thermodynamic limit of the ratchet:** decide whether net matter is (a) finite-size fluctuation (mean asymmetry  $\rightarrow 0$  as  $N \rightarrow \infty$ ), (b) spontaneous symmetry selection (final-value distribution bimodal at  $\pm m^*$ ), or (c) true dynamical rectification, by measuring distributions vs. system size, time, and seed number. The present mean final ratio 1.0013 is small; we do not speak of matter dominance without this decision.
4. **Marrying the frame and the register:** the unified geometry of the three-direction metastable structure (the frame on the graph side) and register position; a metric via the chord formula  $R'' = 2R \sin(\Delta\theta/2)$ .
5. **Separating parity and hair onto internal axes:** toward pure position axes plus internal degrees of freedom (the periodic-table program).
6. **E8:** with all instruments null, move the search to the 8-axis spinor structure (a separate theoretical line). Finite-order structure likely lives not in lattice sizes but in operators, internal axes, and spinor closures.
7. **Closed form and antisymmetric structure of the many-body flow:** the two-body vertex had a closed-form exact solution (pointwise rotation). In the mediated adjacency sum, multiple rotation tangents act simultaneously and the individual driving scalars  $s_{ee'} = \text{Im}(\bar{z}_{e'} z_e)$  can change under the other adjacent flows, so the closed form is not trivially inherited. Can the whole many-body vertex be written as a state-dependent antisymmetric flow  $\dot{\mathbf{z}} = A(\mathbf{z})\mathbf{z}$  with  $A^T = -A$ ? If so, the uniqueness theorem advances to the geometric form “closure conservation selects the antisymmetric tangent flow on the many-body edge space.” This is the next theoretical stronghold.

## 15. What This Paper Does Not Claim; Falsifiability

This paper does not claim that the real universe is this model. The two-factor law, seed unification, and register locality are consequences of “if this dynamics and this embedding are assumed.” Falsification conditions: (a) generation that bypasses the condensation gate (equal rates from non-condensed states); (b) non-unified seeds (a divergence of inflation seed and matter seed violating the  $\varepsilon$ -law); (c) the discovery of graph locality (concentration systematically changing generation rates); (d) breakdown of the transplanted laws ( $p \neq 2$  or non-constant  $C$ ) — any of these, demonstrated in a more constrained experimental

system, rejects the embedding claims. The value of the assumption is measured not by correctness but by the **demonstrated connection** — the physics of two independently published papers reproducing simultaneously as different chapters of a single, switchless dynamics.

**Closing:** the entire content of this paper fits in three lines. The unstable growth of the linear part saturates and builds a condensed order (the three-direction metastable structure). That condensed order opens the generation rate of the always-on nonlinear vertex. Hence — how inflation ends is how matter generation begins.

## 16. Reproducibility

We follow the series convention (self-contained reproduction; originals imported read-only after snapshot verification; criteria fixed before execution).

- **Originals:** Paper 8 `run_preliminary_seed_ablation_v1.py` (SHA-256 75a10a5b...); generator `v3 make_parent_white_harmonics_n_only_v3.py` (SHA-256 d3217579...).
- **Code of this paper** (repository folder `universal_interaction_manybody_connection_v1`, under the series folder for dimensional generation structure): the design and theorem documents; stage 1–3 engines and tests (`run_stage1_multislice_engine_v1.py`, `run_stage2_vertex_engine_v1.py`, `run_stage2_tests_corrected_v2.py`, `run_stage3_shared0_v2_and_hair_v1.py`, etc.); discriminator experiments (`run_stage2_discriminators_v1.py`); GENESIS (`run_genesis_v1.py`, `_v2_local`, `_v3_register_local`); the remaining-holes experiments (`run_holes_h1_h2_v1.py`, `run_h3_ladder_front_v1.py`); the ratchet (`run_ratchet_test_v1.py`); motion (`run_kinetic_dispersion_demo_v1.py`, `run_genesis_3d_demo_v1-v2.py`); figure generation (`make_paper_figures_multibody_v1.py` — experiments whose series were not stored are re-run with identical seeds and checked against the stored summary values before plotting).
- **Result JSONs:** every number in the text agrees with the corresponding `*_result_*.json`.
- **Machine verification of the mediated theorem:** `test_mediated_vertex_closure_v1.py` (T1–T5).

## References

### Within the series (dependence indicated)

- [1] N. Kihara, *Generation of Fermionic Structure Is Induced, Autocatalytic, and Pair-Correlated* (2026), Zenodo. Concept DOI: 10.5281/zenodo.21808091 (the embedded vertex and all transplanted laws; throughout)
- [2] N. Kihara, *Geometric Rapid Expansion Is Specific to Unstable Self-Consistent Closures* (2026), Zenodo. Concept DOI: 10.5281/zenodo.21798854 (onset modes, three-direction condensation, the  $\varepsilon$ -law, the generator; Sections 3, 6, 8)
- [3] N. Kihara, *Paper 7: The Temporal Structure of the N-Body Relational Closed Wave System* (2026), Zenodo. Concept DOI: 10.5281/zenodo.21543070 (the canonical phenomena of the linear dynamics; Section 4)
- [4] N. Kihara, *Paper 8: Causal Separation by Two-Stage Seed Removal* (2026), Zenodo. Concept DOI: 10.5281/zenodo.21614402 (the direct original of the linear-dynamics code; Sections 4, 16)
- [5] N. Kihara, *Fundamental Axioms of the Anonymous Equal-Amplitude Composite Wave Model* (2026), Zenodo. Concept DOI: 10.5281/zenodo.21315735 (the two axioms; Section 5)
- [6] N. Kihara, *Future-Phase Position-Acceleration Map in an AB Two-Body Closed Phase System and the Inverse-Square Law via Harmonic Closure* (2026), Zenodo. DOI: 10.5281/zenodo.21441081 (the harmonic-closure dispersion  $\omega_k = k \omega_1$ ; Section 9 only)
- [7] N. Kihara, *The Generative Structure of Fermions* (2026), Zenodo. Concept DOI: 10.5281/zenodo.21766706 (the original definitions of the readout  $\theta$ ,  $R$ , and the fermionic mask; Section 4)

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